Electromagnetic Theory & Quantum Physics

01.02 · Original practice problems

Show your reasoning, use SI units where appropriate, and state any assumptions. Constants are supplied in the questions that need them.

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A MAIS Physics & Astronomy Club study resource. These are original exercises, not official IOAA problems.

PROBLEMS

01 From wavelength to photon

A spectral line has wavelength 500 nm. Use $c = 3.00 \times 10^{8}$ m s^{-1}, $h = 6.63 \times 10^{-34}$ J s and 1 eV $= 1.60 \times 10^{-19}$ J. (a) Find the frequency. (b) Find the energy of one photon in joules and electronvolts. (c) How many such photons carry 1.0 J of energy?

02 Blackbody comparison

Two spherical stars have the same radius. Star A has surface temperature 6000 K and star B has surface temperature 3000 K. Assume blackbody emission and use Wien’s displacement constant $b = 2.90 \times 10^{-3}$ m K. (a) Find each peak wavelength using $\lambda_{max}T = b$. (b) Compare their luminosities. (c) State one reason why a real stellar spectrum is not a perfect blackbody.

03 Radiation at a distance

A spherical star emits isotropically with luminosity 3.8×10^{26} W. (a) Find the radiative flux at a distance of 1.50×10^{11} m. (b) What flux would be measured twice as far away? (c) A perfectly absorbing flat detector of area 0.010 m^2 faces the star at the first distance. Find the incident power, ignoring all intervening absorption.

04 A quantum threshold

A metal has work function 2.0 eV. Light of wavelength 400 nm falls on it. Use $hc = 1240$ eV nm. (a) Find the photon energy and maximum kinetic energy of an emitted electron. (b) Find the threshold wavelength. (c) At a wavelength longer than the threshold, explain what happens if the light intensity is increased while the wavelength is held fixed.